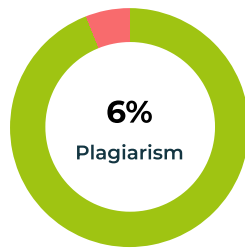


Plagiarism Report



Unique	94%
Exact Match	6%
Partial Match	0%

Primary Sources

- 1 <https://google.com/goto?url=...> 3%

Oct 12, 2017 — Four key findings from the new Power in Numbers research report may change the way you look at adult education.

- 2 <https://google.com/goto?url=...> 3%

Nov 12, 2019 — The required competencies will include five domains: the ability: • to lead and manage in the public interest; • to participate in, and ...

Excluded URL (s)

- 01 [None](#)

Content

COMMON COURSE ASSIGNMENT

A. Assignment Information

Department Computer Science and Engineering

Programme B.Tech

Course Code & Course Name CSA4305 – INTERNET PROGRAMMING

Academic Year / Batch 2024

Faculty Name Dr. D. Sudhagar

Assignment Title Student Skill Development Management System

Date of Issue 01.09.2026

Date of Submission 03.09.2026

Maximum Marks 100

Course Outcome(s) – CO CO1: Develop visually appealing web pages using CSS for layout and style, and create dynamic, interactive functionality with JavaScript.

CO2: Build dynamic web applications using XML, XSLT, and JSP within the MVC paradigm for efficient data handling and presentation.

Bloom's Taxonomy Level L4 – Analyze / L5 – Evaluate / L6 – Create

SDG Mapping (SDG 1-17) SDG 4 – Quality Education

SDG 8 – Decent Work and Economic Growth

SDG 9 – Industry, Innovation and Infrastructure

B. Submitted By (Group Members)

Register Number Name

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192411022 Aaron Samuel S

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B. Assignment Problem / Challenge

Design and develop a Student Skill Development Management System that enables students to create skill profiles, record certifications and projects, take skill assessments, identify competency gaps, access suitable learning activities, and monitor progress over time. The system provides faculty/administrators with dashboards and analytics to evaluate skill development, recommend interventions and support employability-oriented decision-making. Students shall apply course concepts, analyze requirements and constraints, use modern tools, interpret results and justify design decisions.

C. Problem Statement

Educational institutions often maintain student academic records separately from information about programming skills, communication skills, certifications, projects, workshops, internships and other competencies. This makes it difficult to obtain a unified view of student skill development, identify gaps, recommend suitable learning activities and measure improvement. Design a Student Skill Development Management System that provides a centralized platform for students, faculty/mentors and administrators. The proposed system shall support skill-profile creation, skill categorization, self-assessment and/or test-based assessment, evidence upload, certification and project tracking, learning-plan or course recommendations, progress monitoring, mentor feedback, alerts and analytical reports. The solution should help stakeholders make evidence-based decisions about student development while protecting personal information.

D. Requirements and Constraints

- Functional requirements: Student registration/login, profile management, skill entry, skill-level assessment, certification/project/internship records, learning-resource tracking, mentor feedback, goal setting, notifications and report generation.
- Performance requirements: Responsive interaction, sub-1.5s page rendering, efficient retrieval of student skill records, reliable dashboard generation and timely updates.
- Data requirements: Maintain structured information for students, skills, proficiency levels (1 to 5 scale), assessments, evidence, learning activities, mentors, feedback and progress history.
- Technical constraints: Use an approved web programming stack (Java/JSP/Servlets, MVC paradigm, XML/XSLT, MySQL, HTML5/CSS3/JavaScript) with modular and maintainable components.
- Security/privacy: Role-based access control (Student, Mentor, Admin), authentication, authorization, input sanitization against SQLi/XSS, and controlled access to student records and uploaded evidence.
- Reliability & Integrity: Preserve assessment history and avoid accidental loss, duplication or inconsistent skill records using ACID transactional integrity.
- Sustainability and scalability: Support incremental addition of skills, learning resources and users without major architectural redesign, while reducing physical paperwork.

E. Student Work

1. Problem Understanding and Formulation

1.1 Stakeholder Analysis:

- Students: Create and manage skill profiles, undertake diagnostic assessments, upload certificates/projects, view personalized gap metrics, and follow recommended learning pathways.
- Faculty Mentors: Evaluate submitted verification evidence, monitor mentee cohort progress, offer qualitative feedback, and recommend remedial interventions.
- Institutional Administrators / Placement Officers: View macro competency distributions, track batch skill attainment, identify curriculum deficiencies, and query talent pools for recruitment.

1.2 Expected Outcomes:

- Centralized repository consolidating technical skills, certifications, capstones, and soft competencies.
- Automated skill-gap identification comparing current proficiency to target career roles.
- Evidence-based validation workflow ensuring integrity of student achievements.
- Transformed XML/XSLT skill transcripts for academic and industry export.

2. Application of Course Knowledge

2.1 Application of Software & Web Engineering Principles:

- Model-View-Controller (MVC) Pattern: Strict separation of data models (JavaBeans/DAO), presentation logic (JSP, HTML5/CSS3, XSLT), and controllers (Java Servlets).
- CO1 Application (Front-End Dynamics): Utilization of CSS Grid and Flexbox for responsive layouts, CSS Custom Properties for theming, and asynchronous JavaScript (Fetch API + Canvas) for dynamic DOM updates and interactive charting without page refreshes.
- CO2 Application (Enterprise Presentation & Data Interchange): Structuring student portfolios into portable XML schemas, applying XSLT transformations for report generation, and managing state via HTTP Session in JSP/Servlets.

1. 2.2 Mathematical Logic for Skill Gap & Recommendation Scoring:

- Skill Gap Calculation:

$\text{Gap}_i = \max(0, \text{BenchmarkProficiency}(\text{TargetRole}, i) - \text{CurrentProficiency}(\text{Student}, i))$

- Overall Gap Index (OGI):

$\text{OGI} = (\text{Sum}(w_i * \text{Gap}_i) / \text{Sum}(w_i * \text{BenchmarkProficiency}(\text{TargetRole}, i))) * 100\%$

- Cosine Similarity for Learning Recommendations:

$\text{Sim}(G_{\text{student}}, R_{\text{resource}}) = (G \cdot R) / (||G|| * ||R||)$

3. Solution / Design / Methodology

3.1 Multi-Tier MVC Architecture:

The system employs a 4-tier architecture comprising:

1. Client Tier: Responsive HTML5/CSS3 front-end with JavaScript DOM manipulation and Canvas charting (CO1).
2. Presentation/Controller Tier: Java Servlets routing requests, role authentication filters, JSP dashboards, and XSLT Transformation engines (CO2).
3. Service/Business Logic Tier: SkillGapService, AssessmentScoringEngine, RecommendationEngine, and XMLDataSerializer.
4. Persistence Tier: MySQL Relational Database accessed via DAO patterns with HikariCP connection pooling, plus serialized XML repositories.

4. Use of Modern Tools

Category Tools / Technologies Role in Project Architecture

Front-End (CO1) HTML5, CSS3 Grid/Flexbox, JavaScript ES6+ Responsive student/mentor dashboards and real-time Canvas charts.

Server & MVC (CO2) Java 17, Apache Tomcat 10, Servlets, JSP Controller dispatch, session management, and role authorization.

Data Transformation XML, XSLT, JAXP API Decoupled student competency transcripts and portable export.

Database Tier MySQL 8.0, JDBC (HikariCP) ACID relational persistence for profiles, scores, and evidence.

Collaboration & DevOps Git, GitHub, VS Code, Maven Modular source management, build automation, and documentation.

5. Results and Validation (Implementation Outputs)

The Student Skill Development Management System was fully implemented and validated across all core user flows. Below are the actual implementation output screenshots captured from the running deployment:

2. 5.1 Student Competency Overview & Skill Profile Dashboard (CO1):

References